Trainings

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

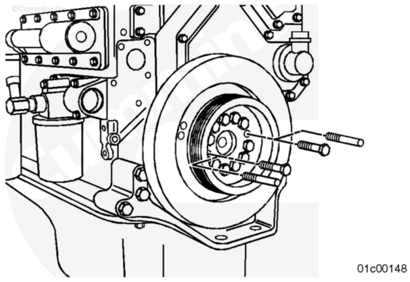
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the cooling fan and water pump drive belt. Refer to Procedure 008-003 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-003.html)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/377/377-013-005.html)
- Remove the cooling fan and water pump belt tensioner. Refer to Procedure 008-080 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-080.html)

Remove

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

component or assembly.



⚠ CAUTION ⚠

Do not use a hammer or a screwdriver to remove the vibration damper(s). These tools can damage the vibration damper(s).

Remove two of the vibration damper and crankshaft pulley retaining capscrews.

Install two guide studs into the holes.

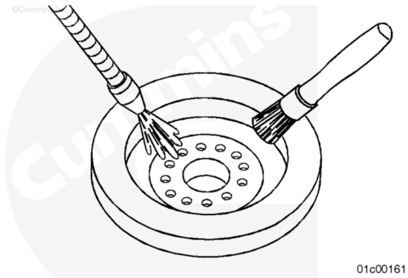
Remove the remaining capscrews.

Remove the pulley and vibration damper.

Clean and Inspect for Reuse

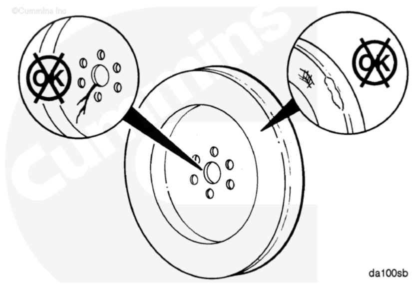
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the vibration damper(s) with a solvent cleaner.

Always replace the vibration damper(s) if the crankshaft nose has been damaged. Crankshaft nose breakages are primarily caused by excessive torsional activity.



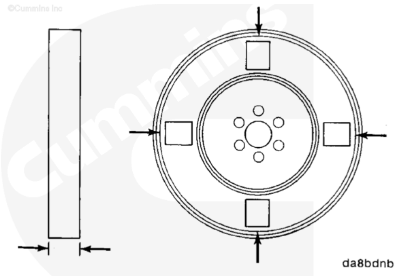
The vibration damper **must always** be measured after a malfunction involving damaged gears in the front gear train.

Check the mounting web for cracks.

Check the housing for dents or raised surfaces.

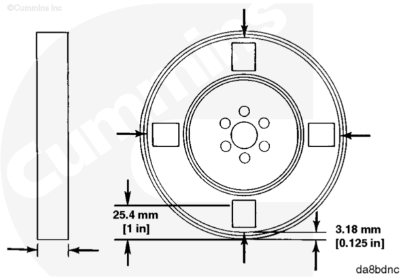
The vibration damper **must** be replaced if the mounting web is cracked or if the housing has dents or raised surfaces.

Remove the paint on both sides of the vibration damper in four locations, spaced 90 degrees apart.



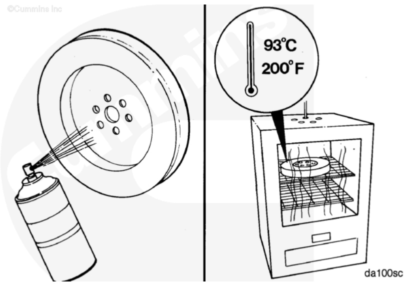
Measure and record the thickness at two points at each of the four locations.

The difference between any two of the eight measurements **must not** exceed 0.25 mm [0.010 in].



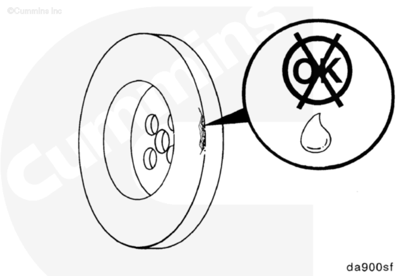
Spray the vibration damper with spot check developer, Type SKD-NF, or equivalent.

Heat the vibration damper in an oven (rolled lip side down) at 93°C [200°F] for 2 hours.



⚠ WARNING ⚠

Wear protective gloves when handling parts that have been heated to reduce the possibility of personal injury.



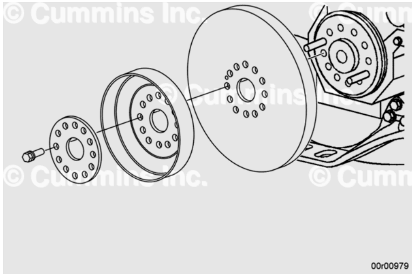
Remove the vibration damper from the oven and check for fluid leakage.

If there is leakage, replace the vibration damper.

Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

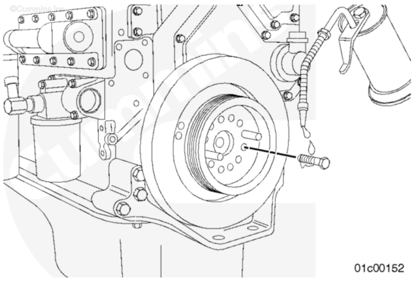


Make sure the mounting surfaces of the crankshaft nose, vibration damper, and the pulley are clean, dry, and free from burrs.

Install two guide studs into the crankshaft nose.

Align the dowel pin in the crankshaft with the dowel hole in the vibration damper.

Install the vibration damper, pulley, and clamping plate onto the guide plate.



Use clean engine oil to lubricate the capscrew threads.

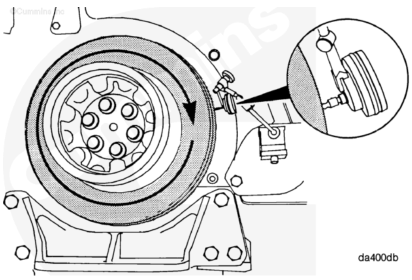
Install ten capscrews.

Remove the two guide studs and install the remaining capscrews.

Tighten the capscrews in a star pattern.

Torque Value: 175 n•m [129 ft-lb]

Eccentricity Check



Clean the outside surface of the vibration damper.

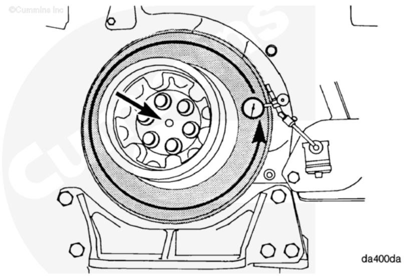
To measure the vibration damper eccentricity (out of round), install a dial indicator onto the gear cover as illustrated.

Rotate the crankshaft with the accessory driveshaft one complete revolution (360 degrees), and record the total indicator movement.

Vibration Damper Eccentricity		
mm		in
0.28	MAX	0.011

If the vibration damper is **not** within specifications, it **must** be replaced.

Wobble Check



To measure wobble (face alignment), install a dial indicator as illustrated.

Push the crankshaft to the front or rear of the engine and set the indicator to “0” (zero).

Rotate the crankshaft one complete revolution (360 degrees) while maintaining the position of the crankshaft, either toward the front or rear of the engine.

Record the total indicator movement.

Vibration Damper Wobble Measurement		
mm		in
0.28	MAX	0.011

If the vibration damper is **not** within specifications, it **must** be replaced.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fan and water pump belt tensioner. Refer to Procedure 008-080 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-080.html)
- Install the fan and water pump drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/377/377-008-002.html)
- Install alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/377/377-013-005.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for proper operation.

Last Modified: 15-Jan-2021